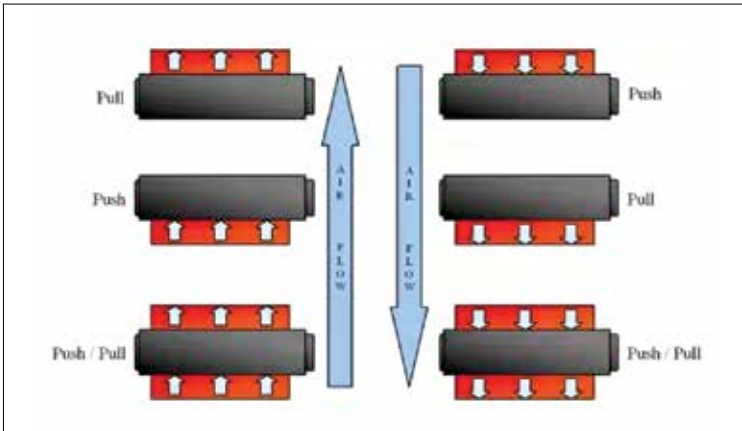


The material used in the fans also matter. Since fans spinning at high RPM are under immense stress thanks to a multitude of forces acting upon it, one of the being centripetal force, the fan blades undergo temporary distortion in the form of thermal expansion or flexing. If the material used is not rigid enough, then the distortion is greater and the fan blades might collide with the edge and break off. There are two ways of tackling this, either increase rigidity or reduce the blade length so that there more room for expansion. The latter affects cooling performance, so you need to balance the two. In order to increase rigidity, brands experiment with different materials. Noctua is well-known for their new Sterrox liquid-crystal polymer which allows them to prevent excessive expansion while ensuring a fair amount of rigidity. Moreover, the material helps reduce resonance and vibration as well.



Push-pull configurations.

The curvature and edges of the individual fans blades are also important to ensure smooth airflow. Noctua's fans have channels cut into the outer edges of the blades to allow for reduced turbulence and thus, a more silent operation. A lot of the brands have tiny microstructures which help cut down wind anomalies so that they don't affect the air flow in a harmful manner. Lastly, fans should have some sort of damping material at the mounting points. Most brands now have a layer of rubber for this.

When there are multiple fans connected, you need to worry about the orientation of the fans with respect to each other. How the fans are oriented